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DEPARTMENT OF CSE(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

DATABASE MANAGEMENT SYSTEMS LABORATORY

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EXERCISE –8: PROCEDURES **Date:10/09/2026**

1. Procedure to Find Factorial of a Given Number

AIM

To write and execute a **PL/SQL procedure** to find the factorial of a given number.

ALGORITHM

1. Start.
2. Read the number N.
3. Initialize FACT to 1.
4. Initialize I to 1.
5. Repeat while I <= N:
    - FACT = FACT × I
    - Increment I by 1.
6. Display FACT.
7. Stop.

PL/SQL PROCEDURE

```
SET SERVEROUTPUT ON;


CREATE OR REPLACE PROCEDURE FACT(N IN NUMBER)
IS
   FACT NUMBER := 1;
   I    NUMBER := 1;
BEGIN
   WHILE I <= N LOOP
      FACT := FACT * I;
      I := I + 1;
```

```
    END LOOP;
    DBMS_OUTPUT.PUT_LINE('Factorial of ' || N || ' = ' || FACT);
END;
/
```

EXECUTION

```
EXEC FACT(5);
```

OUTPUT

```
Factorial of 5 = 120
```

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2. Procedure to Find Sum of N Natural Numbers

AIM

To write and execute a **PL/SQL procedure** to find the sum of the first N natural numbers.

ALGORITHM

1. Start.
2. Read the number N.
3. Initialize SUM to 0.
4. Initialize I to 1.
5. Repeat while I <= N:
    - SUM = SUM + I
    - Increment I by 1.
6. Display SUM.
7. Stop.

PL/SQL PROCEDURE

```
SET SERVEROUTPUT ON;

CREATE OR REPLACE PROCEDURE SUM_N(N IN NUMBER)
IS
    SUM_VALUE NUMBER := 0;
    I         NUMBER := 1;
BEGIN
    WHILE I <= N LOOP
        SUM_VALUE := SUM_VALUE + I;
        I := I + 1;
```

```
    END LOOP;

    DBMS_OUTPUT.PUT_LINE('Sum of first ' || N ||
                    ' natural numbers = ' || SUM_VALUE);
END;
/
```

EXECUTION

```
EXEC SUM_N(10);
```

OUTPUT

```
Sum of first 10 natural numbers = 55
```

---

3. Procedure to Find Fibonacci Series

AIM

To write and execute a **PL/SQL procedure** to generate the Fibonacci series for N terms.

ALGORITHM

1. Start.
2. Read the number of terms N.
3. Initialize:
    - A = 0
    - B = 1
4. Display A and B as the first two terms.
5. Calculate C = A + B.
6. Assign A = B.
7. Assign B = C.
8. Repeat steps 5–7 until N terms are generated.
9. Stop.

PL/SQL PROCEDURE

```
SET SERVEROUTPUT ON;

CREATE OR REPLACE PROCEDURE FIBONACCI(N IN NUMBER)
IS
   A NUMBER := 0;
```

```
    B NUMBER := 1;
    C NUMBER;
    I NUMBER := 1;
BEGIN
    DBMS_OUTPUT.PUT_LINE('Fibonacci Series:');

    WHILE I <= N LOOP
        DBMS_OUTPUT.PUT(A || ' ');

        C := A + B;
        A := B;
        B := C;

        I := I + 1;
    END LOOP;

    DBMS_OUTPUT.NEW_LINE;
END;
/
```

EXECUTION

```
EXEC FIBONACCI(10);
```

OUTPUT

```
Fibonacci Series:
0 1 1 2 3 5 8 13 21 34
```

---

4. Procedure to Check Whether a Number is Perfect or Not

AIM

To write and execute a **PL/SQL procedure** to check whether the given number is a **perfect number** or not.

Definition

A **perfect number** is a number whose sum of its proper divisors is equal to the number itself.

For example:

6 → Divisors: 1, 2, 3

1 + 2 + 3 = 6

Therefore, 6 is a Perfect Number.

ALGORITHM

1. Start.
2. Read the number N.
3. Initialize SUM to 0.
4. Initialize I to 1.
5. Repeat while I < N.
6. Check whether N is divisible by I.
7. If N MOD I = 0, add I to SUM.
8. Increment I by 1.
9. Compare SUM with N.
10. If SUM = N, display **Perfect Number**.
11. Otherwise, display **Not a Perfect Number**.
12. Stop.

PL/SQL PROCEDURE

```
SET SERVEROUTPUT ON;


CREATE OR REPLACE PROCEDURE PERFECT_NUMBER(N IN NUMBER)
IS
   SUM_VALUE NUMBER := 0;
   I        NUMBER := 1;
BEGIN
   WHILE I < N LOOP
      IF MOD(N, I) = 0 THEN
         SUM_VALUE := SUM_VALUE + I;
      END IF;


      I := I + 1;
   END LOOP;
```

```
    IF SUM_VALUE = N THEN
        DBMS_OUTPUT.PUT_LINE(N || ' is a Perfect Number');
    ELSE
        DBMS_OUTPUT.PUT_LINE(N || ' is Not a Perfect Number');
    END IF;
END;
/
```

EXECUTION

```
EXEC PERFECT_NUMBER(6);
```

OUTPUT

6 is a Perfect Number

For another example:

```
EXEC PERFECT_NUMBER(10);
```

Output:

10 is Not a Perfect Number

Common perfect numbers: 6, 28, 496, 8128

Complete Exercise–8 Summary

No.	Procedure Name	Example	Output
1	FACT	FACT(5)	120
2	SUM_N	SUM_N(10)	55
3	FIBONACCI	FIBONACCI(10)	0 1 1 2 3 5 8 13 21 34
4	PERFECT_NUMBER	PERFECT_NUMBER(6)	Perfect Number

Important for all four procedures: after END;, keep / on the next line. Then use EXEC procedure_name(value);